Seasonal ARIMA
Looking a Season Back

Seasonal ARIMA — using recent values and last year



Advanced Financial Time Series

We review ARIMA, then add a new ingredient: seasonality

Winter is coming… — stationarity → AR · MA → ARIMA → seasonal differencing → seasonal ARIMA

00	Review	**Stationarity · AR · MA · ARIMA** Constant mean and variance · lags · error lags
01	Seasonality	**A season ago informs today** Seasonal data · seasonal differencing
02	Notation	**Little letters and big letters** (p, d, q)(P, D, Q)S
03	Numbers to Equations	**ARIMA(1,0,1)(2,1,0)12** Differences first · the rest · forecasting
04	Practice	**Choose and run** Season length · order selection · Python

Little letters are non-seasonal, big letters are seasonal — the same idea used twice

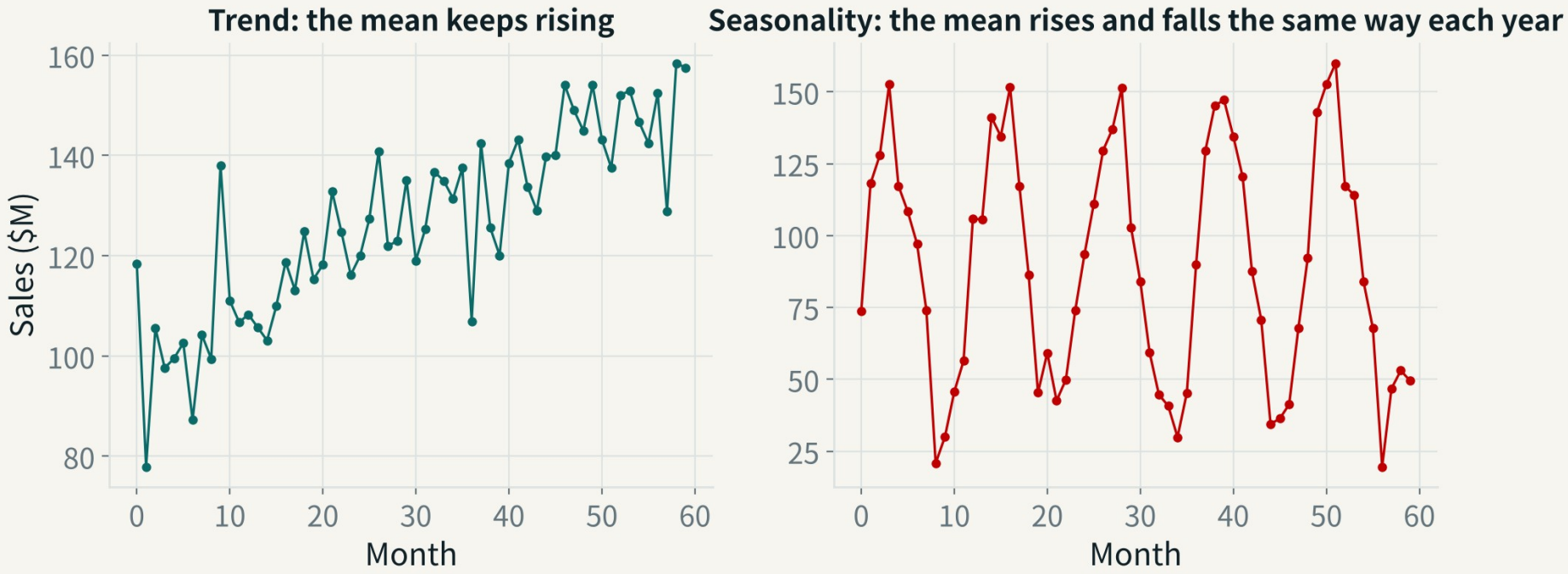
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ARIMA Review

Can't believe all these videos tie together…

A trend or seasonality changes the mean, so the series is not stationary

Consistency of mean and variance — ARIMA needs a stationary series (constant mean and variance)



Changes in mean — trending, seasonality? NOT stationary

AR forecasts with past values, MA with past errors

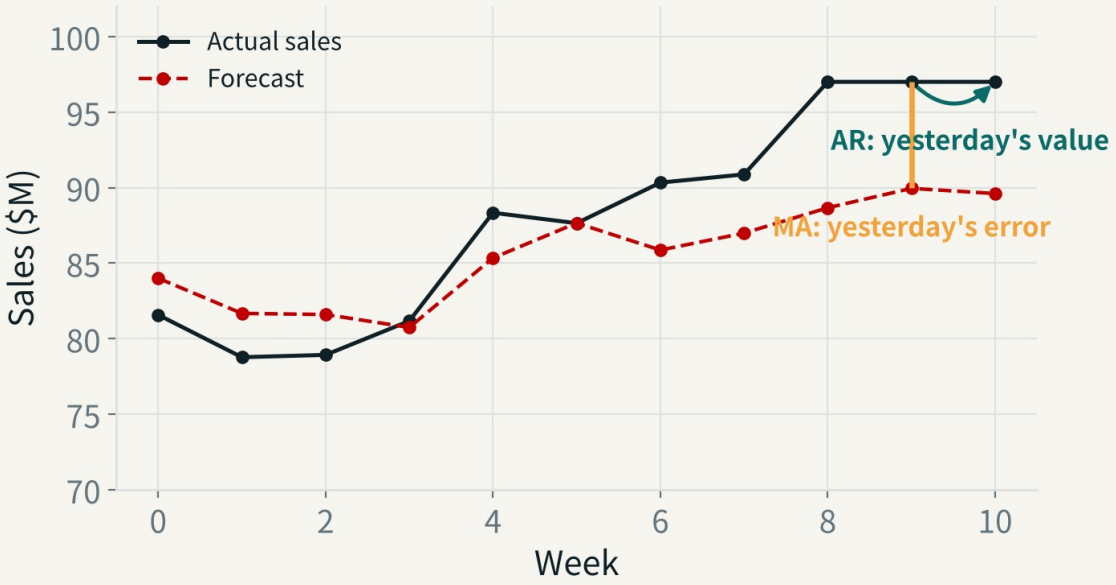
Remember AR and MA models? — black: actual sales, red dashed: forecast

AR: lags · MA: error lags

$$\text{AR}(p): \; Y_t = \omega + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + e_t$$
$$\text{MA}(q): \; Y_t = \omega + \theta_1 e_{t-1} + \cdots + \theta_q e_{t-q} + e_t$$

How to read

- AR: predicts today from yesterday's value
- MA: corrects by yesterday's miss



Two ingredients — past values (lags) and past errors (error lags)

ARIMA differences the series to stationarity, then uses AR and MA together

ARIMA models are typically written as the following

ARIMA(p, d, q)

$$\text{ARIMA}(\underbrace{p}_{\#\,\text{AR terms}}, \underbrace{d}_{\#\,\text{first differences}}, \underbrace{q}_{\#\,\text{MA terms}})$$

When d = 1

$$d = 1: \quad \Delta Y_t = Y_t - Y_{t-1} \text{ is modeled with } \text{AR}(p) + \text{MA}(q)$$

How to read

- p: number of AR terms
- d: number of first differences
- q: number of MA terms
- d = 1: model ΔY instead of Y

ARIMA(p, d, q) = d differences + p AR terms + q MA terms

01

Seasonality

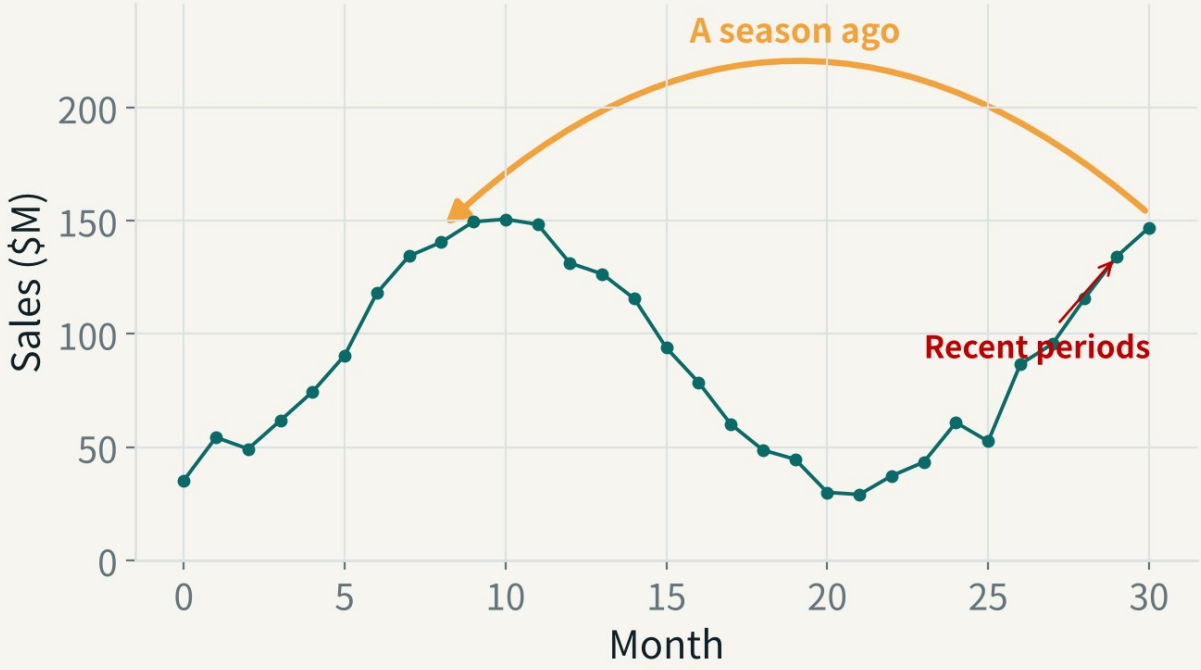
Uh oh… different data — a season ago matters, not just recent periods

In seasonal data, a season ago informs today as well as recent values

What if your data had a seasonal nature to it? — this April looks like last April

How to read

- Red: recent periods (usual AR · MA)
- Amber: one season (12 months) ago
- Both are used to forecast



Not just recent time periods, but also a season ago

Differencing removes both trend and seasonality

Common (not only) solution — differencing

First difference: removes trend

$$\Delta Y_t = Y_t - Y_{t-1} \quad \text{(trend)}$$

Seasonal difference: removes season

$$\Delta_S Y_t = Y_t - Y_{t-S} \quad \text{(season: same point last season)}$$

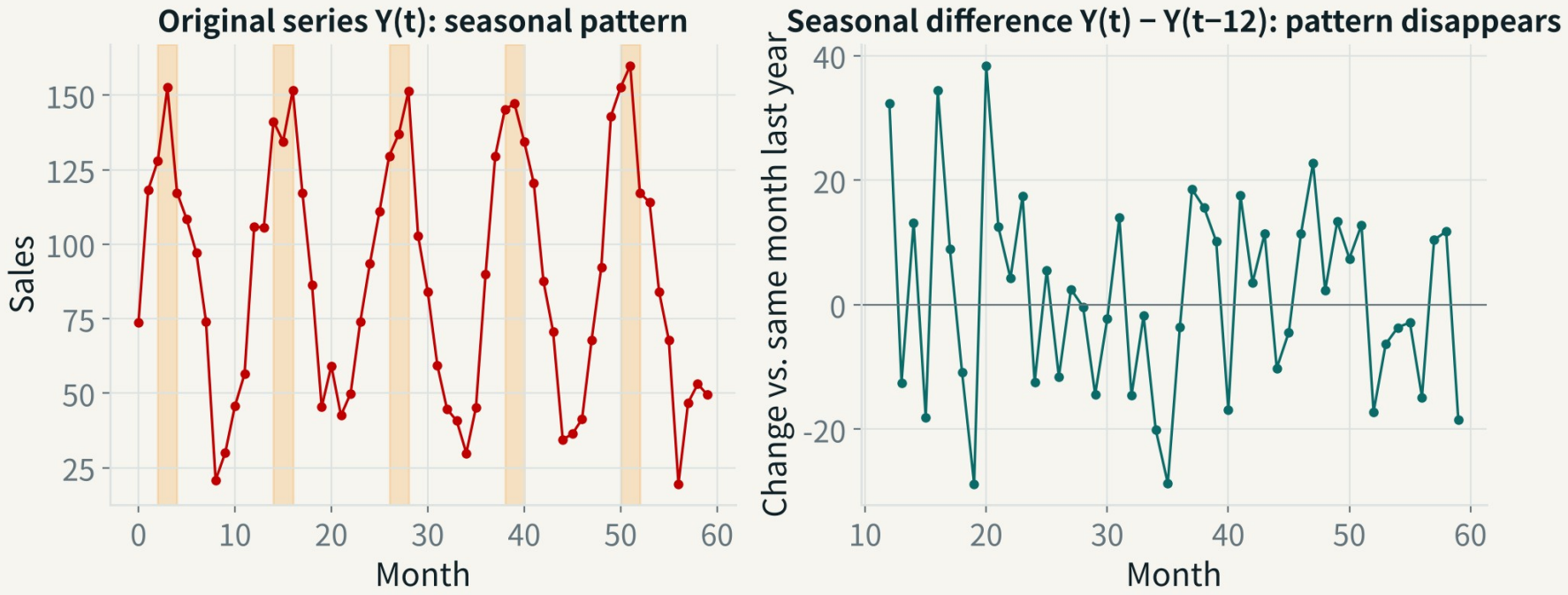
How to read

- S: length of a season (12 for monthly)
- April 2025 − April 2024
- = change vs. same month last year
- Use both differences if needed

Seasonal difference = current value minus the same point last season

A 12-month seasonal difference removes the repeating pattern

Left: simulated monthly sales (amber: yearly peak season), right: Y(t) − Y(t−12)



Fit AR · MA to what remains after seasonal differencing

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Seasonal ARIMA Notation

That's a LOT of letters — little letters and big letters

Seasonal ARIMA is written with two brackets and a season length S

Seasonal ARIMA models are typically written as the following

ARIMA(p, d, q)(P, D, Q)S

$$\text{ARIMA}(\underbrace{p, d, q}_{\text{non-seasonal}})(\underbrace{P, D, Q}_{\text{seasonal}})\underbrace{S}_{\text{season length}}$$

How to read

- First bracket: non-seasonal (recent)
- Second bracket: seasonal (per season)
- S: season length
- Also called SARIMA

Little letters are non-seasonal, big letters are seasonal

Little and big letters do the same job at different spacings

Little letters look back one period at a time, big letters one season (S periods) at a time

	Non-seasonal (little)	Seasonal (big, S = 12)
AR terms	**p: Y(t−1), Y(t−2), …**	**P: Y(t−12), Y(t−24), …**
Differences	**d: Y(t) − Y(t−1)**	**D: Y(t) − Y(t−12)**
MA terms	**q: e(t−1), e(t−2), …**	**Q: e(t−12), e(t−24), …**
Season length	**—**	**S = 12 (monthly) · 4 (quarterly)**

Every big-letter lag is a multiple of S — 12, 24, 36, …

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Numbers to Equations

Writing out ARIMA(1,0,1)(2,1,0)12 line by line

Step 1: Difference first — D = 1, S = 12 means change from last year

Differences first! — in ARIMA(1,0,1)(2,1,0)12, d = 0 and D = 1

One seasonal difference

$$\underbrace{D = 1,\ S = 12}_{\text{one seasonal difference}} \quad : \qquad W_t = Y_t - Y_{t-12}$$

How to read

- W(t): the seasonally differenced series
- d = 0: no first difference
- Now build the model on W(t)

Handle the differencing orders (d, D) first, then read the rest on W(t)

Step 2: W(t) depends on 1 month ago, 1 and 2 years ago, and the last error

Rest of the numbers! — p = 1, q = 1, P = 2, Q = 0 (intuitive reading)

Orders → which past values matter

$$W_t \leftarrow \underbrace{W_{t-1}}_{p=1:\ \phi_1},\ \underbrace{W_{t-12},\ W_{t-24}}_{P=2:\ \Phi_1,\ \Phi_2},\ \underbrace{\varepsilon_{t-1}}_{q=1:\ \theta_1} \qquad (d=0,\ Q=0)$$

How to read

- Lowercase ϕ, θ: non-seasonal
- Uppercase Φ: seasonal AR
- Φ1 → 12 months ago
- Φ2 → 24 months ago

Caution: standard SARIMA multiplies the two ARs — the true equation adds interaction lags

The backshift operator B shifts a series back by one period

Backshift operator B — the same as the lag operator L in econometrics; only the letter differs by field

Definition and numerical example

Definition

$$B\,Y_t = Y_{t-1}, \qquad B^2 Y_t = B(B\,Y_t) = Y_{t-2}$$
$$B^k Y_t = Y_{t-k}, \qquad B\,c = c \ \ (\text{constant})$$

Example $Y_1, Y_2, Y_3, Y_4 = 100,\ 104,\ 110,\ 107$

$$B\,Y_4 = Y_3 = 110, \qquad B^2 Y_4 = Y_2 = 104$$
$$(1 - B)\,Y_4 = 107 - 110 = -3$$

How to read

- B: statistics / Box–Jenkins
- L: econometrics
- B = L: same meaning and rules
- Shifts the time, not the value
- A constant stays the same

Applying B once = taking the value one period earlier; k times = k periods earlier

With B, differencing, AR, and MA become a single polynomial in B

A polynomial ϕ(B) acts like a function applied to the series — powers of B add their exponents

Common forms and the multiplication rule

$$\Delta Y_t = (1 - B)\, Y_t = Y_t - Y_{t-1}$$

$$\Delta_{12} Y_t = (1 - B^{12})\, Y_t = Y_t - Y_{t-12}$$

$$\mathrm{AR}(1): \quad Y_t - \phi Y_{t-1} = (1 - \phi B)\, Y_t = e_t$$

$$\mathrm{MA}(1): \quad Y_t = (1 + \theta B)\, e_t$$

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p \;\Rightarrow\; \mathrm{AR}(p): \quad \phi(B)\, Y_t = e_t$$

$$B^a B^b = B^{a+b}: \quad (1 - B)(1 - B^{12}) = 1 - B - B^{12} + B^{13}$$

How to read

- 1 − B: first difference
- $1 - B^{12}$: seasonal difference
- ϕ(B): AR polynomial
- $B \cdot B^{12} \rightarrow$ 13 periods back

Multiplying brackets adds lags — the reason lags 13 and 25 appear in multiplicative SARIMA

Standard SARIMA multiplies the non-seasonal AR and the seasonal AR

Multiplicative SARIMA — each bracket is a polynomial in the backshift operator B

Exact form of SARIMA(1,0,1)×(2,1,0)12

$$\underbrace{(1-\phi_1 B)}_{\text{non-seasonal AR}}\ \underbrace{(1-\Phi_1 B^{12}-\Phi_2 B^{24})}_{\text{seasonal AR}}\ W_t = \underbrace{(1+\theta_1 B)}_{\text{MA}}\ \varepsilon_t$$

$$W_t = (1-B^{12})\,Y_t = Y_t - Y_{t-12}$$

How to read

- The brackets are multiplied
- Hence the '×' notation
- Multiplying creates extra terms

Multiplying the two brackets creates new terms at 13 and 25 periods back

Expanding the product adds interaction terms at lags 13 and 25

13 = 12 + 1, 25 = 24 + 1 — non-seasonal AR (lag 1) meets seasonal AR (lags 12, 24); constant omitted

Expanded form

$$W_t = \phi_1 W_{t-1} + \Phi_1 W_{t-12} + \Phi_2 W_{t-24} \underbrace{- \phi_1 \Phi_1 W_{t-13} - \phi_1 \Phi_2 W_{t-25}}_{\text{interaction lags}} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

How to read

- Products of ϕ1 and Φ — no new estimates
- ϕ1 0.5, Φ1 0.4 → lag 13: −0.20
- Φ2 0.2 → lag 25: −0.10

SARIMA ≠ AR with only lags 1 · 12 · 24 — interaction lags 13 · 25 always come along

In plain words, the two rules count last year's previous month twice

Example: forecasting this July — the monthly rule (ϕ) looks at June, the seasonal rule (Φ) at last July

① Monthly rule · ϕ

This month ≈ last month

- This July resembles this June
- This June carries part of last June

② Seasonal rule · Φ

This month ≈ last year

- This July resembles last July
- Last July also carries part of last June

③ Remove the overlap

Last June = 13 months ago

- It enters both rules
- Subtract the overlap ϕΦ once
- $0.5 \times 0.4 \rightarrow$ coefficient -0.2

Both rules contain last year's previous month, so remove the overlap at lag 13 — same for lag 25

Filtering season and month in turn creates the interaction term

Why lag 13 — de-seasonalizing last month subtracts its value a year earlier (13 months ago)

Applying the two filters in turn (P = 1)

$$
\begin{aligned}
&(1)\ \text{seasonal filter:} \quad U_t = W_t - \Phi\, W_{t-12} \\
&(2)\ \text{monthly filter:} \quad U_t - \phi\, U_{t-1} = \varepsilon_t, \qquad U_{t-1} = W_{t-1} - \Phi\, W_{t-13} \\
&\Rightarrow\ (W_t - \Phi W_{t-12}) - \phi\,(W_{t-1} - \Phi W_{t-13}) = \varepsilon_t \\
&\Rightarrow\ W_t = \phi W_{t-1} + \Phi W_{t-12} \underbrace{- \phi\Phi\, W_{t-13}}_{\text{interaction}} + \varepsilon_t
\end{aligned}
$$

How to read

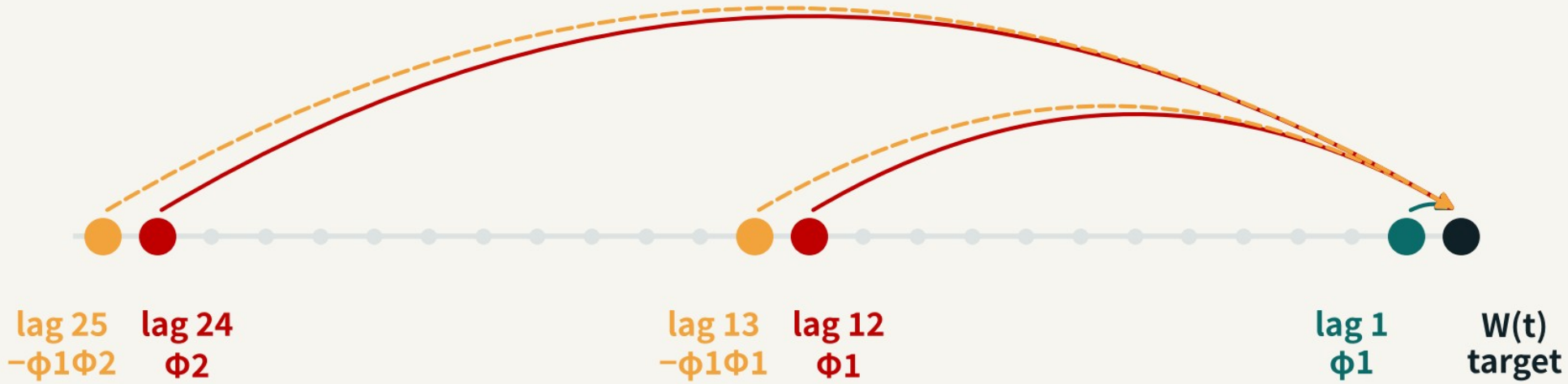
- ϕ + Φ counts the overlap twice
- Subtract ϕΦ at lag 13
- ϕ 0.5, Φ 0.4 → −0.2

The interaction term corrects double counting — its coefficient is not estimated separately

On a timeline, lags 1, 12, 13, 24, and 25 all connect to W(t)

SARIMA(1,0,1)×(2,1,0)12 — positions of the past values that explain W(t) = Y(t) − Y(t−12)

Red/green: lags set directly by the orders Orange dashed: interaction lags from multiplying the two ARs (13 = 12 + 1, 25 = 24 + 1)



Last month + same month last year + the month before that (lag 13)

Forecast W first, then add back last year's value to return to Y

Undoing the difference — since W(t+1) = Y(t+1) − Y(t−11), Y(t+1) = Y(t−11) + W(t+1)

One-month-ahead forecast (with interaction terms)

$$\hat{Y}_{t+1} = Y_{t-11} + \hat{W}_{t+1}$$
$$\hat{W}_{t+1} = \hat{\phi}_1 W_t + \hat{\Phi}_1 W_{t-11} + \hat{\Phi}_2 W_{t-23} - \hat{\phi}_1\hat{\Phi}_1 W_{t-12} - \hat{\phi}_1\hat{\Phi}_2 W_{t-24} + \hat{\theta}_1 \varepsilon_t$$

How to read

- Last year + forecast change
- Estimated by MLE
- Longer horizons: recursion

SARIMA forecast = same month last year + forecast of this year's change

04

Practice

Set the season length, choose the orders, and run

The season length S comes from the data's cycle

How many periods until the same pattern repeats?

Common values of S

$$S = 12 \quad (\text{monthly})$$
$$S = 4 \quad (\text{quarterly})$$
$$S = 7 \quad (\text{daily data, weekly cycle})$$

How to read

- Monthly sales: 12
- Quarterly GDP: 4
- Daily visitors: 7
- Trading-day weekly cycle: 5
- Daily financial returns: weak seasonality

A wrong S makes the seasonal terms look at the wrong past

Set differencing first, then pick the other orders by AIC·BIC

Practical order — usually d, D ≤ 1 and P, Q ≤ 2 are enough

Step	Task	Criterion
1	Set S	Repeating cycle in plots and autocorrelation
2	Set D (usually 0 or 1)	D = 1 if a seasonal pattern remains
3	Set d (usually 0 or 1)	d = 1 if a trend remains, ADF test
4	Choose p, q, P, Q	Minimum AIC·BIC (auto_arima)
5	Check residuals	No autocorrelation left in residuals

Python: SARIMAX(y, order=(1,0,1), seasonal_order=(2,1,0,12)).fit()

05

Summary

ARIMA that uses both recent values and a season ago

SARIMA(1,0,1)×(2,1,0)12 describes the whole model in six numbers

Reading the numbers in the brackets

Number	Letter	Meaning	In the equation
1	p	One non-seasonal AR term	$\phi 1 \cdot W(t-1)$
0	d	No first difference	—
1	q	One non-seasonal MA term	$\theta 1 \cdot e(t-1)$
2	P	Two seasonal AR terms	$\Phi 1 \cdot W(t-12), \Phi 2 \cdot W(t-24)$
1	D	One seasonal difference	$W(t) = Y(t) - Y(t-12)$
0	Q	No seasonal MA	—

※ Multiplicative, so interaction terms $-\phi 1 \Phi 1 \cdot W(t-13)$ and $-\phi 1 \Phi 2 \cdot W(t-25)$ are included

Little letters: next door, big letters: a season ago — difference first, multiply for lags 13·25

Not stationary? Difference it

Seasonal? Subtract a season ago

Little + big letters = seasonal ARIMA

Winter is coming…
Seasons always come back